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# Odoo Migration Cookbook

*Version 19 to Version 20*

COMMUNITY EDITION AND ENTERPRISE EDITION · FIRST EDITION · SEPTEMBER 2026

Mr. Mustufa Rangwala
Ms. Sananaz Mansuri

Ahmedabad, Gujarat, India
Website: https://www.probuse.com
Website: https://probuseappdemo.com/
Email: contact@probuse.com

This cookbook lists the code replacements required when moving custom modules from Odoo 19.0 to Odoo 20.0. It covers **Odoo Community** and **Odoo Enterprise**. Each entry is a 19 → 20 change: a model, field, method, view xpath, QWeb directive, HTTP type, or JavaScript API that 19 accepted and 20 does not.

The replacements are taken from the official Odoo **19.0** and **20.0** source trees ( `odoo/odoo` Community, including `base` and the standard addons, and the matching Enterprise addons).

### Applies to

| Mark | Meaning |
|---|---|
| **CE + EE** | Same change in Community 20 and Enterprise 20 |
| **CE** | Community 20 only, or a Community model that Enterprise does not use |
| **EE** | Enterprise 20 only (Payroll, Budget, Field Service, Sign, Enterprise Helpdesk, Enterprise Accounting extras) |

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### Authors

Mr. Mustufa Rangwala
Ms. Sananaz Mansuri



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Website: https://probuseappdemo.com/
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Confirm each replacement against the Odoo 20.0 Community or Enterprise source you deploy.

## About the publisher

Probuse Consulting Service Pvt. Ltd. is an Odoo consulting and product company based in Ahmedabad, Gujarat, India, founded on 1 November 2014. The firm implements, customizes, and supports Odoo for customers worldwide.

Website: https://www.probuse.com
Website: https://probuseappdemo.com/
Email: contact@probuse.com

## About the authors

**Mr. Mustufa Rangwala** is Founder, CEO, and Managing Director of Probuse Consulting Service Pvt. Ltd. He has worked in the Odoo / OpenERP ecosystem since the early versions and previously worked with Odoo India.

**Ms. Sananaz Mansuri** is a Consultant and Odoo techno-functional implementer. She works on requirement analysis, process mapping, configuration, and delivery. She is the GCC contact for Probuse Consulting Service Pvt. Ltd.

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# 1. Security

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### CE + EE

Odoo 20 loads `ir.access.csv`. It does not load `ir.model.access.csv` plus XML `ir.rule` records. XML `<record model="ir.rule">` raises `KeyError: 'ir.rule'`.

### Odoo 19

```
id,name,model_id:id,group_id:id,perm_read,perm_write,perm_create,perm_unlink
access_lock,lock,model_my_lock,base.group_user,1,1,1,0
```

### Odoo 20 — `security/ir.access.csv`

```
id,name,model_id,group_id/id,operation,domain
access_my_lock,my.lock,my.lock,base.group_user,cru,
```

| Column | Odoo 20 |
|---|---|
| `model_id` | Technical model name (`sale.order`). Not `module.model_x`. |
| `group_id/id` | Group XML id. Empty = global rule. |
| `operation` | `crud`, `cru`, `cr`, … — not `rc`, and not `perm_read` columns |
| `domain` | Record-rule domain, or empty |

Further 19 → 20 rules on this file:

- `model_id/id` from 19 is not `model_id`. The 20 column needs the technical name.
- Domains `[(1, '=', 1)]` and `[('1', '=', 1)]` fail validation. Use an empty domain.

- A domain on a non-stored field raises `Field has no SQL representation`. Use a stored field or a stored path.
- Company domain `('company_id', 'in', company_ids + [False])` fails. Use:

```
['|', ('company_id', '=', False), ('company_id', 'in', company_ids)]
```

- Custom `res.groups` XML must be listed in the manifest **before** `ir.access.csv` if the CSV references those groups.

---

# 2. ORM and Python

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**CE + EE** unless marked otherwise.

| Odoo 19 | Odoo 20 |
| --- | --- |
| `_sql_constraints = [('name', 'sql', 'msg')]` | `_name = models.Constraint(` |
| `self.fields_view_get(...)` | `self._get_view(...)` |
| `self.check_access_rights('read')` | `self.check_access('read')` |
| `self.check_access_rights('read', raise_exception=False)` | `self.has_access('read')` (b |
| `self.user_has_groups('sales_team.group_sale_manager')` | `self.env.user.has_groups('` |
| `self._context` | `self.env.context` |
| `self.env.user.company_id` | `self.env.company` |
| `ir.config_parameter.get_param` / `set_param` | `get_int` / `get_str` / `get_bo` `set_str` / `set_bool` / `set_f` |
| `pytz` | `datetime.UTC` and `zoneinfo` |
| window action `toggle_active` | `action_archive` / `action_ur` |
| write `archive=True` | write `active=False` |
| `mail.thread.cc` | `mail.thread` |

```
# 19
_sql_constraints = [
    ('code_uniq', 'unique(code, company_id)', 'Code must be unique per company.'),
]

# 20
_code_uniq = models.Constraint(
    'unique(code, company_id)',
    'Code must be unique per company.',
)
```

```
# 19
self.check_access_rights('read')

# 20
self.check_access('read')
self.has_access('read')
```

```
# 20 — selection_add requires ondelete
state = fields.Selection(selection_add=[
    ('extra', 'Extra'),
], ondelete={'extra': 'set default'})
```

`_name` listed in `_inherit` requires that parent model to exist in the registry. **CE** a module that `_name = 'helpdesk.sla'` and also `_inherit = 'helpdesk.sla'` fails unless **EE** `helpdesk` is installed. **CE-only** graphs use `_name` only. **EE** graphs that load Enterprise Helpdesk keep `_inherit`.

`mail.thread.cc` is removed. A model that inherits `rating.mixin` already inherits `mail.thread`. Listing both on 20 raises an MRO error.

```
# 20
_inherit = ['portal.mixin', 'mail.activity.mixin', 'rating.mixin']
```

Two Many2many fields on the same model to the same comodel, without `relation=`, collide on 20. Set `relation=` on at least one.

A 20 compute must assign every record. A 20 compute must not `return` a value dict. Stored and non-stored fields cannot share one compute method.

`method=True` on a computed field is invalid. `tracking=` on a related field requires `mail.thread`. Related Selection must not re-declare `selection` / `default` / `tracking`. `placeholder=` is XML-only.

A required SQL column without `default=` raises `NOT NULL` on create. `required=True` on a new Many2one cannot add `NOT NULL` when the table already has rows.

`self.env['some.model']` is a recordset. `_inherit` is on `self.env.registry['some.model']`.

### COPY, WRITE ORDER, AND TRANSLATIONS

A Char field named `name` now defaults to `copy=mark_as_copy('name')`. Duplicating a record appends "(copy)" to that value. Override `copy=` when the 19 behaviour must stay a plain copy.

```
from odoo.tools.translate import mark_as_copy

name = fields.Char(copy=mark_as_copy('name'))
code = fields.Char(copy=False)
```

`create()` and `write()` run field inverses in `(field.write_sequence, field definition order)`. Custom inverses that assumed 19 order can break. Set `write_sequence=` when two inverses must run in a fixed sequence.

```
partner_id = fields.Many2one('res.partner', write_sequence=10)
```

Reading a One2many or Many2many returns **only records the user can access**. Code that counted or mapped the full relational cache of forbidden ids will see a shorter recordset.

Module translations are stored in **that module's PO**. 20 no longer dumps them into `base`. Export and import translations per module.

### WEBSITE ON THE ENVIRONMENT

CE + EE when `website` is installed. 19 model code often used `request.website`. 20 Environment exposes `env.website` from `context['website_id']`. Prefer `self.env.website` in models. `request` is not bound in report parsers and several ORM contexts (Chapter 4).

```
# 20
website = self.env.website
```

---

# 3. Search and domains

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## CE + EE

| Odoo 19 | Odoo 20 |
| --- | --- |
| `_name_search` | `_rec_names_search` and/or `_search_display_name` |
| `from odoo.osv import expression` / `expression.AND` / `expression.OR` | `from odoo.fields import Domain` / `Domain.AND` / `Domain.OR` |
| `from odoo.fields import Domain, AND, FALSE_DOMAIN` | `from odoo.fields import Domain` only ( `Domain.FALSE` ) |
| `_name_search(..., args=..., name_get_uid=...)` | `(self, name, domain=None, operator='ilike', limit=None, order=None)` |
| `_read_group_stage_ids(self, stages, domain, order)` | `_read_group_stage_ids(self, stages, domain)` |
| view modifier `in 'x'` / `not in 'x'` | `== 'x'` / `!= 'x'` (tuple `in ('a', 'b')` is valid) |

```
# 20
class AccountAnalyticAccount(models.Model):
    _inherit = 'account.analytic.account'
    _rec_names_search = ('name', 'code', 'custom_number', 'partner_id')
```

```
from odoo.fields import Domain

def _search_display_name(self, *args, **kwargs):
    return Domain(super()._search_display_name(*args, **kwargs))
```

`odoo.tools.Query` is not exported from `odoo.tools`. Import `SQL` from `odoo.tools`.

### DOMAIN OPERATOR `ACCESS`

20 adds the domain operator `access`. It is valid on `id` and on Many2one fields. The value is `'read'`, `'write'`, `'create'`, or `'unlink'`. Use it when a domain must keep only records (or related records) the user may operate on. It does not work on properties.

```
# 20 — current user can write the partner
Domain('partner_id', 'access', 'write')

# 20 — current user can read this record
Domain('id', 'access', 'read')
```

This operator is useful in `ir.access.csv` domains (Chapter 1) when the rule must follow comodel rights.

---

# 4. Binary, Excel, PDF, and time

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### CE + EE

#### BINARY

On the record, a Binary field is `BinaryValue` (`odoo.tools.binary`). `.content` is already raw bytes. `base64.decodebytes(self.datas)` double-decodes or raises `Incorrect padding`.

| Direction | Odoo 20 |
|---|---|
| Read | `value.content` or `bytes(value)` |
| Filename | `value.filename` — optional name on `BinaryValue` / `BinaryBytes` |
| Write | `from odoo.tools.binary import BinaryBytes` — not raw `bytes`, not `base64.b64encode(...)` |
| Image write | `image_apply_opt(pil_image, 'PNG')` then `BinaryBytes(...)`. `image_to_base64` is removed |
| `ir.attachment` | field `raw`, not `datas`; `res_id` is an integer |
| XML binary | `type="bytes"` |
| RPC / HTTP | binary payload is `BinaryValue`, not a bare base64 string |

`BinaryBytes(data, filename=name)` keeps the original file name on the value. RPC and HTTP that still send a raw base64 string without that wrapper fail the 20 type check.

```
from odoo.tools.binary import BinaryBytes
from odoo.tools.image import image_apply_opt

record.image_1920 = BinaryBytes(image_apply_opt(pil_image, 'PNG'))
record.datas = BinaryBytes(stream.getvalue(), filename=name)
```

```
<!-- 19 -->
<img t-attf-src="#{doc.image_1920}"/>

<!-- 20 -->
<img t-att-src="image_data_uri(doc.image_1920) if doc.image_1920 else '/web/static
```

### EXCEL

| Odoo 19 | Odoo 20 |
| --- | --- |
| `xlwt` | `xlsxwriter` (in the 20 virtualenv). Write to `io.BytesIO` with `{'in_memory': True}`. Assign with `BinaryBytes` |
| `xlrd.open_workbook` on `.xlsx` | xlrd 2.0.1 cannot open Office Open XML. Use `openpyxl` for `PK` `.xlsx`. Use xlrd only for OLE `.xls` |

### PDF

| Odoo 19 | Odoo 20 |
| --- | --- |
| `import PyPDF2` | `from odoo.tools.pdf import PdfFileWriter, PdfFileReader` |
| `content_disposition` in `odoo.tools` / `odoo.http` old path | `odoo.http.stream` |
| `_render_qweb_pdf(template_xmlid)` | `_render_qweb_pdf(action_xmlid)` |
| `request.env` in a report parser | `env` — `request.env` raises `RuntimeError: request not bound` |

### TIME

```
from datetime import UTC
from zoneinfo import ZoneInfo

local_tz = ZoneInfo(self.env.user.tz or 'UTC')  # ZoneInfo('utc') fails
if utc_dt.tzinfo is None:
    utc_dt = utc_dt.replace(tzinfo=UTC)
local_dt = utc_dt.astimezone(local_tz)
```

```
resource._attendance_intervals_batch(
    start_dt, end_dt,
    resources_per_tz=resource._get_resources_per_tz(),
)
```

| Odoo 19 | Odoo 20 |
| --- | --- |
| `time_type` on leave / calendar leave | `count_as='absence'` |
| `resource.calendar` field `tz` | removed |
| calendar attendance `name` / `day_period` | removed |

---

# 5. Configuration, HTTP, and mail

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**CE + EE** unless marked **EE**.

| Odoo 19 | Odoo 20 |
| --- | --- |
| `@http.route(..., type='json')` | `@http.route(..., type='jsonrpc')` |
| `ir.actions.server mail_post_method` | `state="mail_post"` + `template_id` + optional `mail_post_autofollow` |
| EE server action `use_in_ai` | `ai_tool_name` matching `[a-zA-Z0-9_]{1,64}` |
| settings inherit `group[@name='preferences']` | `//page[@name='preferences']//group[@name='other_prefe` |
| employee settings xpath | after `employees_setting_container` |

A mail template that inherits an Enterprise model must use the **defining** module's model xmlid. Example **EE**: `account_budget.model_budget_analytic`. Inherit does not create `your_module.model_budget_analytic`.

### `MAIL.TRACKING.VALUE`

**CE + EE.** 19 stores `mail.tracking.value` in core `mail`. 20 moves the model to the Community addon **`mail_tracking`** (Discuss Tracking). Core `mail` still generates tracking messages; it does not keep the stored tracking rows unless `mail_tracking` is installed.

| Odoo 19 | Odoo 20 |
| --- | --- |
| `mail.tracking.value` in `mail` | install `mail_tracking`; then `_name = 'mail.tracking.value'` |
| search `mail.tracking.value` with only `mail` | empty / missing model unless `mail_tracking` is in the graph |

Depend `mail_tracking` when custom code reads stored old/new values.

#### EXTERNAL RPC

`/json/2/<model>/<method>` (JSON-2, `type='json2'`, bearer auth) is the documented external API on 20. The same controller already exists on 19. XML-RPC ( `/xmlrpc/2` ) is still present on both versions. New integrations should call JSON-2. Do not treat XML-RPC as removed in 20.

```
POST /json/2/res.partner/search_read
```

Controller routes that the web client calls stay `type='jsonrpc'` (table above). That is not the same as JSON-2.

---

# 6. SQL reports

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**CE + EE**

### `_TABLE_QUERY` → `_TABLE_SQL`

19 SQL views set `_auto = False` and define `_table_query` (attribute or method). 20 **removes** `_table_query`. The lint on 20 is: overwrite `_table_sql`. Return `odoo.tools.SQL`.

```
# 19
_auto = False

def _table_query(self):
    return self._query()

# 20
from odoo.tools import SQL

_auto = False

def _table_sql(self) -> SQL:
    return SQL("(%s)", self._query())
```

`sale.report` on 20 is the reference: `_table_sql()` builds a `Query` and returns `query.subselect(...)`. A module that only migrated `_select()` / `_group_by()` to `SQL` still fails if it leaves `_table_query` in place.

### `_SELECT()` AND `_GROUP_BY()`

`_select()` and `_group_by()` return `odoo.tools.SQL`. A Python string cannot be concatenated onto that value. `LiteralSQL` has no `.code`.

```
from odoo.tools import SQL

def _select(self):
    return SQL("%s,\n t.custom_type", super()._select())
```

| Report | Odoo 20 hook |
|---|---|
| `report.project.task.user` | `_select()` / `_group_by()` → `SQL` |
| `timesheets.analysis.report` | `_select()` → `SQL` |
| `sale.report` | `_select_dict` + `_groupby_list` |
| `purchase.report` | `_select_list(table)` + `_groupby_list(table)` |
| `account.move` invoice report | `_select_list` |
| `report.pos.order` | wrap `_select` only |

A field added on `analytic.view_account_analytic_line_filter` must also be added on `timesheets.analysis.report` through `_select()`.

Translated `name` in SQL:

```
COALESCE(project_task_type.name->>'en_US', project_task_type.name::text)
```

---

# 7. Views

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CE + EE unless marked otherwise.

### KANBAN

| Odoo 19 inherit | Odoo 20 inherit |
| --- | --- |
| `project.view_project_kanban` body | `project.view_project_card` |
| `project.view_task_kanban` footer | `project.view_task_card` `//t[@t-name='card']//footer` |
| kanban template name | `t-name="card"` |

`project.view_task_kanban` uses `card_id="%(project.view_task_card)d"`.

### RIBBONS AND BUTTON ICONS

```
<!-- 19 -->
<button icon="fa-check"/>
<widget name="web_ribbon" text="Archived" bg_color="bg-info"/>

<!-- 20 -->
<button icon="check"/>
<widget name="web_ribbon" title="Archived" bg_color="text-bg-info" invisible="acti
```

| Odoo 19 | Odoo 20 |
| --- | --- |
| `web_ribbon` `text=` | `title=` |
| `bg_color="bg-info"` / `bg-primary` | `bg_color="text-bg-info"` / `text-bg-primary` / `text-bg-success` / `text-bg-warning` / `text-bg-danger` |
| button `icon="fa-check"` | `icon="check"` — names from `web/icons.py` `ICONS` |
| Font Awesome / GTK icon names on ViewButton | do not draw. `iconFromString` renders `oi` + `data-icon=` |
| Font Awesome on website snippets | Material Symbols ligature (same names as ViewButton) |

Legal ViewButton names include `check`, `save`, `close`, `history`, `edit_square`, `task`, `group`, `book`, `description`, `stethoscope`.

Owl 3 `DropdownItem` takes `attrs=`. `href=` and `dataset=` are invalid props.

### XPATH

View inherit cannot use attribute `string` or `title` as a selector. Use `hasclass(...)`, `name`, `id`, or `t-call`.

Xpath cannot enter the body of a `t-call`. Inherit the called template, or xpath the `t-call` node.

`[@class='X']` is an exact match. Convert the inherit only when the 20 parent class string no longer equals `X`. Use `hasclass('X')` in that case.

Invoice, sale, and purchase order line `product_id` / `name` are inside `<column name="product_and_description">`. `/list/field[@name='product_id']` does not match. Xpath a direct list child (`quantity`, `product_qty`, `analytic_distribution`, `account_id`).

| Form | Odoo 19 xpath | Odoo 20 |
|---|---|---|
| Analytic account | `//header` | no header — insert `<header>` w |
| Timesheet | add `button_box` | `div[@name='button_box']` alre |
| Calendar event | `button_box` | no `button_box` — buttons in `/` `//sheet/widget[@name='web_r` |
| Employee | `oe_title` | `//h1` or `//h1[hasclass('o_em` |
| Job | `user_id` | after `department_id` ; form alre |
| CRM | `//div[hasclass('oe_title')]/div` | `//div[hasclass('oe_title')]` |
| Maintenance request | `archive_equipment_request` / `user_id` | `//header` ; assignees `user_ids` |
| Invoice totals | `@class='oe_subtotal_footer'` | `//group[hasclass('oe_subtot` |
| Equipment | 19 `location_id` | `owner_user_id` |
| Partner list | `complete_name` | `display_name` |
| Partner search | `group[@name='group_by']` | `//search/group` |
| User preferences | `group[@name='preferences']` | `//page[@name='preferences']` |
| Product | `product.product_variant_easy_edit_view` | removed |
| Product action | `product.product_normal_action` | `product.product_variant_act` |
| Sale config menu | `sale.prod_config_main` | `sale.menu_sale_config` |
| Project form | `group[@name='extra_settings']` | `//page[@name='settings']` |
| Timesheet kanban footer | `footer[@class='mt-0 pt-0']` | `div[@name='ts_footer']` |
| eLearning channel | `//field[@name='active']` | no `active` — xpath `name` |

| Form | Odoo 19 xpath | Odoo 20 |
| --- | --- | --- |
| Activity popup | `company_id` | after `summary` |
| POS session | `move_id` | `user_id` |
| POS receipt | Owl `ReceiptHeader` / `pos-receipt-contact` | `point_of_sale.pos_order_rec` |
| Sale form | `show_update_fpos` | `fiscal_position_id` |
| Sale portal sidebar | `navspy` | `div#sale_order_sidebar_butt` |

`product.product_variant_action` context `active_id` is defined only on a product form. A custom `ir.actions.act_window` copied from that action must not keep `active_id` if it opens from another model.

---

# 8. QWeb and reports

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### CE + EE

`<act_window>` and `<report>` are not in Odoo 20 `import_xml.rng`. Use `ir.actions.act_window` and `ir.actions.report` records. `ir.actions.report` has no `report_file`.

```
<record id="action_print_doc" model="ir.actions.report">
    <field name="name">Document</field>
    <field name="model">sale.order</field>
    <field name="report_type">qweb-pdf</field>
    <field name="report_name">my_module.report_document</field>
    <field name="binding_model_id" ref="sale.model_sale_order"/>
    <field name="binding_type">report</field>
</record>
```

| Odoo 19 | Odoo 20 |
| --- | --- |
| `t-esc` | `t-out` |
| `t-raw` | `t-out` |
| `t-field-options` / `t-esc-options` | `t-options` |
| `t-field` | still valid |
| invoice `o.type` | `o.move_type` |
| BoM overview header `BoM Cost` | `Total Cost` |
| `from odoo.tools import html2plaintext` | `from odoo.tools.mail import html2plaintext` |
| `web_editor` Python `get_video_embed_code` | package removed; embed is JS `html_editor` |

An xpath that selected `@t-esc` must select `@t-out`.

### REPORT ENGINES

**CE + EE.** 20 splits PDF rendering into two Community modules that 19 did not ship as addons:

| Engine | Odoo 20 module |
| --- | --- |
| wkhtmltopdf | `base_report_wkhtmltox` |
| Paper Muncher (in-house) | `base_report_paper_muncher` |

A custom report that assumed "one wkhtml pipeline inside `web`" must depend on the engine actually installed. Paper Muncher is the 20 in-house renderer (`https://odoo.github.io/paper-muncher/`).

**EE** If the report bound to `hr.work.entry` rows: those document records are gone. Bind to `hr.payslip` and iterate `worked_days_line_ids`.

---

# 9. Owl 3

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### CE + EE

Odoo 20 web is Owl 3.

| Odoo 19 | Odoo 20 |
|---|---|
| `import { useState } from "@odoo/owl"` | `import { proxy } from "@odoo/owl"` then `this.state = proxy({ ... })` |
| template `state.foo` / `kpis.foo` / `dashboardData.foo` | `this.state.foo` / `this.kpis.foo` / `this.dashboardData.foo` / `this.props.foo` |
| `t-esc` | `t-out` |
| `t-model` on a `proxy()` field | `t-model` accepts a `signal` only. `proxy()` fields use `t-att-value` + `t-on-input` |
| `t-set` inside `t-call` as a sub-template variable | slot body — pass parameters as attributes on `<t t-call="..."/>` |
| `standardActionServiceProps` from `action_service` | `@web/webclient/actions/action_plugin` |
| Chart.js v2 options | Chart.js v4 ( `cutout` , `plugins.legend` , `plugins.title` , `scales.y.beginAtZero` ) at `web/static/lib/Chart/Chart.js` |
| `_attachment_format` | `_store_attachment_fields` |
| `mail.AttachmentList` props `attachments` | `attachmentGroups` ; container `o-mail-AttachmentContainer` |
| `mail.assets_messaging` | empty — put JS/XML in `web.assets_backend` |
| `window.alert` in systray | notifications; company is `user.activeCompany` |

```
import { proxy } from "@odoo/owl";
this.state = proxy({ total: 0 });
```

```
<t t-out="this.state.total"/>
<t t-out="this.kpis.open"/>
```

`t-as` loop locals stay unprefixed. A static CSS class is not a template identifier ( `ud-widget` stays `ud-widget` ).

```
static props = { "*": true };
```

```
import { signal } from "@odoo/owl";
this.searchTerm = signal("");
```

Mail store **CE + EE**:

```
Store().add(channel, "_store_channel_fields")._build_result()
```

```
this.store["mail.thread"].getOrFetch({ model: "discuss.channel", id })
```

A Python method called from Owl as `[[], payload]` must be `@api.model` .

Owl 3 also drops or relocates several 19 hooks. Replace them when the 20 component fails to compile or never re-renders.

| Odoo 19 | Odoo 20 |
| --- | --- |
| `this.render()` | the `proxy` / `signal` update re-renders; do not call `render` |
| `t-portal` | removed — render into the target node, or a 20 portal helper if the standard component still exposes one |
| `onWillUpdateProps` / `onWillRender` / `onRendered` | removed from Owl 3. Use `onWillStart` , `onMounted` , `onWillUnmount` , and `onPatched` |

# 10. Accounting and payments

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**CE** + **EE** unless marked **EE**.

| Odoo 19 | Odoo 20 |
| --- | --- |
| invoice lines `journal_line_ids` | `line_ids` |
| invoice salesperson group | `sale_info_group` (customer invoices) |
| vendor bill extra fields | `accounting_info_group` |
| journal entry other info | `other_tab_entry` |
| header dates / flags | `header_right_group` |
| `res.partner.bank.acc_number` | `account_number` |
| `res.partner.bank.acc_holder_name` | `holder_name` |
| `res.partner.bank.bank_id.name` / `bank_id.bic` | `bank_name` / `bank_bic` |
| `account.journal.bank_id` | `bank_name` |
| `res.bank` | removed — Char, or `res.partner.bank` |
| `account.analytic.account.group_id` | `plan_id` |
| `payment.provider.state` | `is_live` / `is_published` |
| `payment.controllers.post_processing` | removed |
| payment transaction group `//group//group` | `//group[@name='transaction_details']` |
| invoice report `o.type` | `o.move_type` |
| `account.group` | removed |
| `account.account.group_id` | `parent_id` / `parent_ids` on `account.account` |
| account `code` required | `code` is computed / optional ( `code_store` , `placeholder_code` ) |
| `account.payment` state `in_process` | `paid` |
| `account.payment` state `paid` | `reconciled` |

### ACCOUNT GROUPS → PARENT ACCOUNTS

19 has `account.group` and `account.account.group_id`. 20 deletes `account.group`. Hierarchy lives on `account.account`:

```
# 20
parent_id = fields.Many2one('account.account', string="Parent Account")
parent_ids = fields.Many2many('account.account', compute='_compute_parent_ids')
```

Replace searches on `account.group` and writes to `group_id` with parent accounts. Account codes are no longer a required stored Char; 20 computes `code` (`compute_sql`). Reports that `JOIN` a mandatory `account_account.code` must tolerate empty codes and use `placeholder_code` when displaying.

#### PAYMENT STATES

| Odoo 19 | Odoo 20 |
|---|---|
| `draft` | `draft` |
| `in_process` | `paid` |
| `paid` | `reconciled` |
| `canceled` | `canceled` |

Filters, automations, and domains that still use `in_process` match nothing. A payment that 19 called "Paid" is **Reconciled** on 20. A payment that 19 called "In Process" is **Paid** on 20.

#### FIXED ASSETS — EE

19 Enterprise treats a depreciation template as `account.asset` with `state == 'model'`. 20 Enterprise adds **`account.depreciation.model`**. Bind templates, default depreciation methods, and CoA-generated models to that model. `account.asset` remains the asset document.

`payment.method.provider_id` is required. Create the provider first. There is no `payment.payment_method_card` xmlid in `payment`.

Custom fields copied from payment to the journal entry go in `_generate_move_vals` / `_synchronize_to_moves`.

Invoice line `_compute_account_id` assigns `line.account_id`. It does not `return`.

---

# 11. Sales, purchase, and products

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### CE + EE

| Odoo 19 | Odoo 20 |
| --- | --- |
| `sale.order` state `done` | removed. Valid: `draft` / `sent` / `sale` / `cancel` |
| `purchase.order.line` UoM `product_uom_id` | `uom_id` |
| `sale.order.line` / `account.move.line` UoM | still `product_uom_id` |
| `product.type == 'product'` | `is_storable` |
| `product.pricelist.item._compute_price(..., date)` positional | `date=` keyword (`_compute_base_price` the same) |
| `sale.prod_config_main` | `sale.menu_sale_config` |
| `product.product_normal_action` | `product.product_variant_action` |
| `product.product_variant_easy_edit_view` | removed |

`sale.order.line` still has `product_uom_qty` and `product_uom_id`.

---

# 12. Inventory, MRP, and repair

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### CE + EE

| Odoo 19 | Odoo 20 |
|---|---|
| `stock.scrap` | `stock.move` with `is_scrap=True` |
| scrap `name` | `reference` |
| scrap / move UoM `product_uom` / `product_uom_id` | `uom_id` |
| scrap dest `scrap_location` | `company.scrap_location_id` |
| scrap validate | `action_scrap()` |
| scrap form / action | `stock.view_scrap_move_form` / `stock.stock_scrapped_moves_action` |
| `stock.return.picking` | `stock.picking._create_return()` |
| `stock.quant.package` | `stock.package` |
| `stock_picking_batch` | merged into `stock`; inherit `stock.stock_picking_batch_form` |
| `mrp.bom` / `mrp.bom.line` / `repair.order` UoM | `uom_id` |
| picking move list UoM | `//field[@name='move_ids']/list/field[@name='uom_` |
| `mrp.production.show_produce_all` | removed |
| MO button “Produce All” (separate from Produce) | one `button_mark_done` |

19 manufacturing orders expose `show_produce_all` and two header buttons (Produce vs Produce All). 20 removes that field. Close the order with

`button_mark_done` only. Xpaths and automations that test `show_produce_all` fail.

---

# 13. HR, time off, and payroll

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**CE** + **EE** for employee / time off. **EE** for Payroll.

| Odoo 19 | Odoo 20 |
| --- | --- |
| `hr.contract` | `hr.version` |
| SQL join `hr_contract` | `hr_version` |
| `hr.version.wizard` | `employee.contract_template_id` + `_onchange_contract_template_id()` |
| `hr.leave.type` | `hr.work.entry.type` |
| `hr.leave.holiday_status_id` | `work_entry_type_id` |
| `hr.leave.action_confirm` | removed; `action_approve` remains |
| `hr.leave.holiday_type` | `validation_type` |
| inherit time-off type form | `hr_holidays.hr_work_entry_type_view_for` `support_document` |
| `hr.expense.sheet` | `hr.expense` |
| `hr_work_entry.work_entry_type_attendance` | `hr_work_entry.generic_work_entry_type_a` |
| legal leave default | `hr_work_entry.generic_work_entry_type_l` |
| leave XML on `hr.leave.type` | merge onto `hr.work.entry.type` ( `count_as='absence'` , `time_off_selectabl` |
| depend `hr_work_entry_holidays` only | `hr_holidays` |
| EE `hr.payslip.input.type` | removed; `input_usage_payslip` on the salar |
| EE salary rule `category_id` / `struct_id` | `category_ids` / `struct_ids` `[(6, 0, [ref` |
| EE `payslip.paid_amount` | `payslip._get_basic_salary()` |

| Odoo 19 | Odoo 20 |
| --- | --- |
| EE payment wizard `export_format` `selection_add` | `_get_export_format_selection()` |
| EE pay-run kanban | `hr_payroll.hr_payrun_payslip_kanban_car` |
| EE `hr.work.entry` list documents | removed; print from `hr.payslip` `worked_days_line_ids` |

Search filter `approve` on time off is nested. `//search/filter[@name='approve']` does not match. Use `//filter[@name='approve']`.

Several 20 workflows use state `canceled`, not `cancel`.

---

# 14. Project, timesheets, and portal

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**CE + EE unless marked EE.**

| Odoo 19 | Odoo 20 |
|---|---|
| `project.task.user_id` | `user_ids` |
| access domain `task_id.user_id` | `('task_id.user_ids', 'in', [user.id])` |
| `note.note` | `project.task` (`project_todo`) |
| todo `default_memo` | `default_name` (`project.task` has no `memo`) |
| `project.portal_my_home` | `portal.entry` (`project.portal_projects`, `project.portal_tasks`) |
| portal card `active` | `show_in_portal` |
| `portal_client_category` / `portal_service_category` | removed |
| `portal.portal_sidebar` `//t[@t-set='entries']` | `//div[hasclass('col-lg-3')]` |
| invoice portal download | `//div[hasclass('o_download_pdf')]` |
| portal docs wrap | `div.o_portal_docs` |
| `get_records_pager` | removed |

`_prepare_home_portal_values` counters are merged in `counters()`. Group-restricted cards use `is_config_card` and `_filter_visible_portal_cards`.

### PROJECT PROFITABILITY — CE

19 Community `project.project` implements `_get_profitability_items`, `_get_profitability_values`, `_get_profitability_labels`, and related helpers. Those

methods are **not** on Community 20 `project`. Overrides and tests that call them raise `AttributeError`. Enterprise 20 reports margin and budget through the budget apps (Chapter 17), not through those 19 CE hooks.

**EE Field Service:** `industry_fsm` is not in Enterprise 20. Depend `project` (and `sign` if signatures are used).

| Odoo 19 EE | Odoo 20 |
|---|---|
| `industry_fsm` | `project` |
| `industry_fsm.fsm_menu_root` | `project.menu_main_pm` |
| `industry_fsm.group_fsm_user` | `project.group_project_user` |

`is_fsm` is not a standard 20 field. Custom domains that still use it need a local Boolean on `project.project` / `project.task`. Enterprise Planning (`planning.slot`) is a different model.

---

# 15. CRM, calendar, and mail

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### CE + EE

| Odoo 19 | Odoo 20 |
| --- | --- |
| CRM `//div[hasclass('oe_title')]/div` | `//div[hasclass('oe_title')]` |
| lead `currency_id` (standard) | `company_currency` |
| notebook `page[@name='lead']` | `internal_notes` / `extra` |
| Extra Info `company_id` in `categorization` | `//page[@name='extra']//field[@name='company_id']` ( `group[@name='Misc']` ) |
| graph amount | `type="measure"` |
| calendar `button_box` | removed — see Chapter 7 |
| calendar `page_invitations` | `notes` / `options` |
| `mail.tracking.value` in `mail` | Community addon `mail_tracking` (Chapter 5) |

# 16. Website and eCommerce

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### CE + EE

```
<!-- 19 — 500 on 20 public pages -->
<input t-attf-value="#{user_id.partner_id.email}"/>

<!-- 20 -->
<input t-att-value="request.env.user.partner_id.email if not request.env.user._is_
```

| Odoo 19 | Odoo 20 |
| --- | --- |
| `a[@id='add_to_cart']` | `button[@name='add_to_cart']` |
| add-to-cart wrap | `div[@id='add_to_cart_wrap']` / `website_` `#o_wsale_cta_wrapper` |
| `product._is_add_to_cart_possible()` form | removed |
| `website_sale.product_terms_and_conditions` | removed — xpath after `#o_wsale_product_c` |
| `#wrap.js_sale` | removed |
| `website_sale_wishlist` / `website_sale_comparison` | merged into `website_sale` |
| wishlist / compare xmlids | `website_sale.product_wishlist` , `website` `website_sale.model_product_wishlist` |
| comparison controller | `odoo.addons.website_sale.controllers.c` |
| hide price | `combination_info['hide_price']` |
| category form `sequence` | list only — xpath `parent_id` |
| `res.partner.company_type` | removed — `is_company` |
| `res.partner.mobile` | removed — `phone` |
| `res.users.groups_id` | `group_ids` |
| action context `default_company_type` | `default_is_company` |
| `ir.ui.view.groups_id` | unchanged (view access, not the user field) |
| Font Awesome class on website snippets ( `fa fa-...` ) | Material Symbols ligature / `oi` + `data-icon` |
| `request.website` in a model method | `self.env.website` (Chapter 2) |

Website and snippet markup that still ships Font Awesome class names will not draw on 20. Use the same ViewButton / Material Symbols names as Chapter 7 ( `check` , `close` , …). Public pages that read `request` from a model or report parser hit `request not bound` ; use `env.website` .

```
# 19
vals['company_type'] = 'person'
vals['mobile'] = '...'
user_vals['groups_id'] = [(6, 0, ids)]

# 20
vals['is_company'] = False
vals['phone'] = '...'
user_vals['group_ids'] = [(6, 0, ids)]
```

---

# 17. Enterprise budget, Field Service, Helpdesk, and Fleet

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### BUDGET — EE

| Odoo 19 Enterprise | Odoo 20 Enterprise |
|---|---|
| `crossovered.budget` | `budget.analytic` |
| `crossovered.budget.lines` | `budget.line` |
| lines field | `budget_line_ids` |
| amount field | `budget_amount` |
| states | `draft` / `confirmed` / `revised` / `done` / `canceled` |
| form | `account_budget.view_budget_analytic_form` |
| action | `account_budget.act_budget_analytic_view` |
| report binding | `account_budget.model_budget_analytic` |
| `committed_amount` | on `account_budget_purchase` |

CE Community 20 has no `budget.analytic`. A Community budget module that still uses its own model is not the Enterprise model and is not retargeted to `account_budget`.

### HELPDESK

EE Enterprise `helpdesk` remains. CE a custom `helpdesk.sla` `_name` must not `_inherit = 'helpdesk.sla'` unless Enterprise Helpdesk is in the registry (Chapter 2).

These 19 Enterprise addons have **no** 20 folder. Depend `project` / `helpdesk` / `documents` instead of the FSM bridges:

| Odoo 19 Enterprise | Odoo 20 |
|---|---|
| `industry_fsm*` | `project` (Chapter 14) |
| `helpdesk_fsm` / `helpdesk_fsm_report` / `helpdesk_fsm_sale` | `helpdesk` |
| `documents_fsm` | `documents` |

#### FIXED ASSETS — EE

See Chapter 10. 20 adds `account.depreciation.model` . 19 templates were `account.asset` with `state == 'model'` .

#### AI ADDONS — EE

Enterprise 20 ships a set of `ai_*` modules. They are new optional apps, not a rename of a 19 module. Do not add them to a 19 graph unless the 20 database actually uses that stack.

#### FLEET — CE + EE

| Odoo 19 | Odoo 20 |
|---|---|
| `fleet.vehicle.log.services.date` | `date_from` / `date_to` |

---

# 18. Symptom index

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| Exception or result | Chapter |
|---|---|
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# 19. Mapping tables

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**MODELS**

| Odoo 19 | Odoo 20 | Edition |
| --- | --- | --- |
| `hr.contract` | `hr.version` | CE + EE |
| `hr.leave.type` | `hr.work.entry.type` | CE + EE |
| `hr.expense.sheet` | `hr.expense` | CE + EE |
| `hr.payslip.input.type` | removed | EE |
| `stock.scrap` | `stock.move` ( `is_scrap=True` ) | CE + EE |
| `stock.return.picking` | `stock.picking._create_return()` | CE + EE |
| `stock.quant.package` | `stock.package` | CE + EE |
| `note.note` | `project.task` | CE + EE |
| `mail.thread.cc` | `mail.thread` | CE + EE |
| `res.bank` | Char or `res.partner.bank` | CE + EE |
| `industry_fsm` | `project` | EE |
| `stock_picking_batch` | `stock` | CE + EE |
| `website_sale_wishlist` | `website_sale` | CE + EE |
| `website_sale_comparison` | `website_sale` | CE + EE |
| `crossovered.budget` | `budget.analytic` | EE |
| `crossovered.budget.lines` | `budget.line` | EE |
| `hr.version.wizard` | removed | CE + EE |

| Odoo 19 | Odoo 20 | Edition |
| --- | --- | --- |
| `account.group` | `account.account` ( `parent_id` ) | CE + EE |
| `account.asset` `state='model'` | `account.depreciation.model` | EE |
| `mail.tracking.value` (in `mail` ) | `mail.tracking.value` (in `mail_tracking` ) | CE + EE |
| `helpdesk_fsm` / `helpdesk_fsm_report` / `helpdesk_fsm_sale` | `helpdesk` | EE |
| `documents_fsm` | `documents` | EE |

### FIELDS

| Model | Odoo 19 | Odoo 20 | Edition |
|---|---|---|---|
| `sale.order` | state `done` | removed | CE + EE |
| `stock.move.line` / `stock.move` / `purchase.order.line` / `mrp.bom` / `repair.order` | `product_uom_id` / `product_uom` | `uom_id` | CE + EE |
| `sale.order.line` / `account.move.line` | `product_uom_id` | unchanged | CE + EE |
| `res.partner.bank` | `acc_number`, `acc_holder_name` | `account_number`, `holder_name` | CE + EE |
| `res.partner` | `company_type`, `mobile` | `is_company`, `phone` | CE + EE |
| `res.users` | `groups_id` | `group_ids` | CE + EE |
| `ir.attachment` | `datas` | `raw` | CE + EE |
| `project.task` | `user_id` | `user_ids` | CE + EE |
| `product.template` | `type == 'product'` | `is_storable` | CE + EE |
| `hr.leave` | `holiday_status_id` | `work_entry_type_id` | CE + EE |
| `hr.leave` | `holiday_type` | `validation_type` | CE + EE |
| `account.analytic.account` | `group_id` | `plan_id` | CE + EE |
| `crm.lead` | `currency_id` | `company_currency` | CE + EE |
| `fleet.vehicle.log.services` | `date` | `date_from` / `date_to` | CE + EE |
| `payment.provider` | `state` | `is_live` / `is_published` | CE + EE |

| Model | Odoo 19 | Odoo 20 | Edition |
| --- | --- | --- | --- |
| `resource.calendar.leaves` | `time_type` | `count_as` | CE + EE |
| `account.move` report | `type` | `move_type` | CE + EE |
| `account.account` | `group_id` | `parent_id` / `parent_ids` | CE + EE |
| `account.payment` | state `in_process` | `paid` | CE + EE |
| `account.payment` | state `paid` | `reconciled` | CE + EE |
| `mrp.production` | `show_produce_all` | removed ( `button_mark_done` ) | CE + EE |
| Binary | bytes / base64 | `BinaryValue` + `.filename` | CE + EE |

### XMLIDS AND VIEWS

| Odoo 19 | Odoo 20 |
| --- | --- |
| `sale.prod_config_main` | `sale.menu_sale_config` |
| `product.product_normal_action` | `product.product_variant_action` |
| `product.product_variant_easy_edit_view` | removed |
| `project.view_project_kanban` body | `project.view_project_card` |
| `project.view_task_kanban` card | `project.view_task_card` |
| `project.portal_my_home` | `portal.entry` |
| `industry_fsm.fsm_menu_root` | `project.menu_main_pm` |
| `hr_work_entry.work_entry_type_attendance` | `generic_work_entry_type_attendance` |
| pay-run kanban | `hr_payroll.hr_payrun_payslip_kanban_car` |
| Enterprise budget form | `account_budget.view_budget_analytic_for` |
| scrap form | `stock.view_scrap_move_form` |
| batch picking form | `stock.stock_picking_batch_form` in `stock` |
| `_table_query` | `_table_sql()` → `SQL` |
| `request.website` in models | `env.website` |
| wkhtml inside `web` | `base_report_wkhtmltox` / `base_report_paper_muncher` |
| JSON-2 `/json/2` | still present (preferred external API); XML-RP present |

**Odoo Migration Cookbook: Version 19 to Version 20**
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Mr. Mustufa Rangwala and Ms. Sananaz Mansuri

Published by Probuse Consulting Service Pvt. Ltd.
419/A, SAKAR IX, Beside Old Reserve Bank of India,
Near City Gold Cinema, Ashram Road,
Ahmedabad 380009, Gujarat, India

Telephone: +91 96011 19434 · +91 78745 43092
Email: contact@probuse.com
Website: https://www.probuse.com
Website: https://probuseappdemo.com/

Middle East and GCC
Contact: Ms. Sananaz Mansuri

Europe
Contact: Mr. Michel Rodriguez